

# How Replicas Are Organised, and What AmberMeta Can Detect

Findings from the deposited simulation corpus

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## 1 Scope

This note accompanies the AmberMeta SOP and answers the standing question of how users organise replicas and whether AmberMeta can recognise them. It reports what was found in the deposited corpus at `/store7/gentile/data/simulations`, what AmberMeta does with each layout today, and what would have to change to cover the cases it declines.

No code was changed. Everything below was measured against AmberMeta v1.1.0 as released.

## 2 The corpus

Of 50 system directories, 19 hold simulation files. 30 are empty placeholders. One, `sys005`, is *unreadable*: its permissions are `drw-rw-r--`, so the execute bit is missing and no user — including its owner — can traverse it. That is a one-line fix at the source (`chmod u+rx`) and worth doing before any further deposits.

### 2.1 What was tested

Five systems were selected to cover every replica convention present. All were scanned read-only; nothing was written into the corpus.

| System | Files | Logs | Steps | Scan   | Layout                       |
|--------|-------|------|-------|--------|------------------------------|
| sys023 | 1044  | 205  | 213   | 6.4 s  | Chunked, no replicas         |
| sys011 | 1280  | 260  | 268   | 1.9 s  | run1..run4 beside the stages |
| sys004 | 5100  | 0    | 107   | 83.5 s | run1..run6 inside each stage |
| sys021 | 5348  | 1091 | 1105  | 27.8 s | 01..05 inside each stage     |
| sys024 | 167   | 30   | 50    | 1.3 s  | L12Y/prod/01..05             |

All five were parsed without error. Two results stand out independently of replicas:

- **sys004 contains no output logs at all.** 5100 files — 2496 restarts, 2496 trajectories, 104 inputs — and not one `.mdout`. A manifest can still be built, but nothing in it is confirmable: no achieved simulation time, no temperature, no completion status, and every junction between runs reported as an unexplained gap. This is the single most consequential gap found in the corpus.
- **sys011 has 260 logs but only 65 inputs.** The runs are recoverable from the logs, but the requested settings for most of them are not on record.

## 3 The four conventions, and what AmberMeta does

| Convention               | Shape                                           | Auto | Manual |
|--------------------------|-------------------------------------------------|------|--------|
| Replicas within stages   | equilibration/run1..run6, production/run1..run8 | yes  | —      |
| Numbered within stages   | equil/01..05, prod/01..05                       | yes  | —      |
| Replicas beside stages   | run1..run4 next to equilibration/equil..equi4   | no   | yes    |
| Id repeated in filenames | L12Y/prod/01/GB1_204_01_000                     | no   | yes    |

Two of the four are recognised with no intervention, and produce correctly tagged members (`run1..run8` for `sys004`; `01..05` for `sys021`). The other two are *declined*, not mis-grouped — and in both the runs are still discovered and still correctly chained. Only the grouping into members is withheld.

### 3.1 Why the two declines happen

Neither is a defect. AmberMeta's layout inference tags nothing when a tree admits more than one reading, on the grounds that a wrong member claim is worse than none.

**sys011** presents two families whose tag sets are disjoint: `equilibration/equi1..equi4`, which varies at the second path segment, and `run1..run4`, which varies at the first. Disjoint families are refused because they are equally consistent with two separate experiments in one directory. Restricting the scan to the replica directories alone resolves it immediately — `discover --pattern 'run[0-9]'` tags `run1..run4` correctly.

**sys024** repeats the replica identifier inside the filenames: `prod/01/GB1_204_01_000`, `prod/02/GB1_204_02_000`. Grouping keys on the set of run *bases* a directory holds, so each replica directory ends up with a base set of its own, no two directories are ever seen as running the same thing, and no cohort with more than one member is ever formed. This one cannot be fixed by narrowing the scan; the identifier has to be picked manually.

### 3.2 The manual route works for both

**Define replicas...** lets the user choose which path segment names the member. On **sys011** choosing the first segment offers `equilibration`, `leap`, `run1`, `run2`, `run3`, `run4`; on **sys024** choosing the third offers the preparation step names alongside `01..05`. In both cases the replicas are present and correct among the options, and accepting only those produces a manifest identical to one from a conforming layout.

The cost is judgement: the picker offers every directory at the chosen level, including preparation directories that are not replicas. A user who accepts all of them gets a manifest asserting that `leap` is a replica.

## 4 Recommendations

1. **Adopt the layout in §3 of the SOP for new work.** Replicas as sibling directories directly inside their stage, identically named across stages, with the identifier in the directory name only. Both auto-detected conventions in the corpus already follow this; it costs nothing to standardise on it.
2. **Treat missing output logs as a deposit blocker.** **sys004** is the case to cite: a 5100-file deposit from which nothing can be confirmed. A one-line check at deposit time (every run directory holding a `.mdout`) would have caught it.
3. **Fix **sys005**'s permissions** and re-run the survey over all 19 populated systems once the SOP is in use.
4. **Consider two contained improvements to the picker**, in order of value:
  - Mark which offered members *look like* replicas — directories whose run-base sets match each other — and which do not, so a user is steered away from accepting `leap` as a member. This is presentational and does not touch the inference rule.
  - Consider whether a directory whose base sets differ only by an embedded copy of its own directory name (the **sys024** shape) can be recognised safely. This is a real change to the membership predicate and would need its own design; the safe-failure property is worth more than the convenience, so it should not be rushed.
5. **Do not weaken the disjoint-families rule.** The **sys011** refusal is the rule working as intended. The procedural workaround is cheap and the failure mode it prevents — silently merging two experiments into one campaign — is not.